Improved Chaining in non-DAG Regions for Long Read Giraffe

UC SANTA CRUZ
Computational Genomics Lab & Platform

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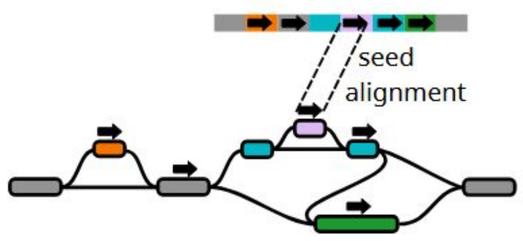
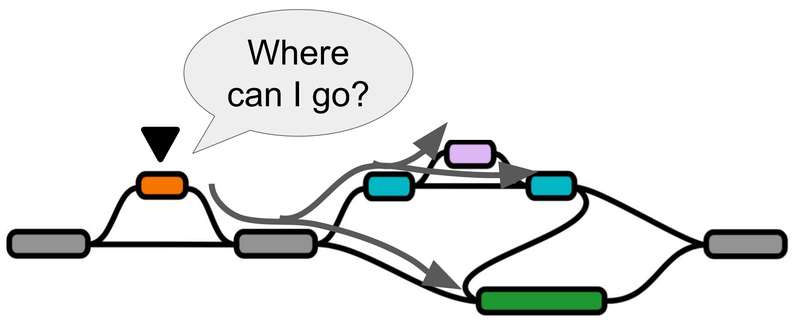
Background

Alignment chaining

Co-linear chain of seeds = alignment "backbone".

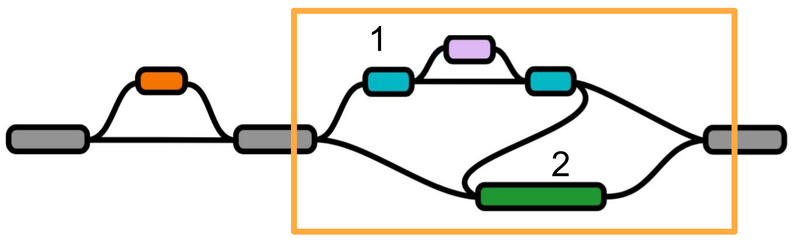


Find distance between seeds in **pangenome graph**.

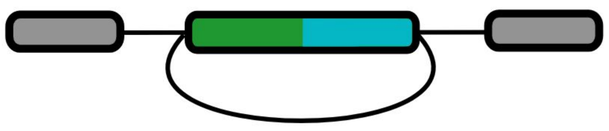


Construct a **zip code tree**, then traverse it to find **graph distances** between pairs of seeds.

Snarls



Want to visit subsections in a **topological order**.

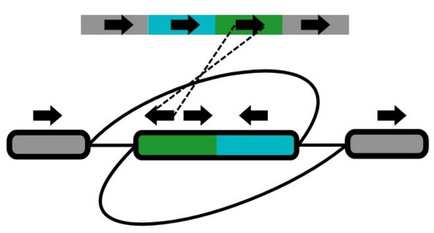


What about **cyclic snarls**?

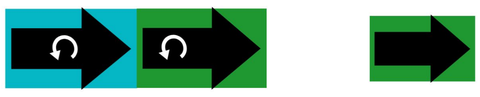
Old heuristics

Artificial linear ordering of cyclic snarl subsections

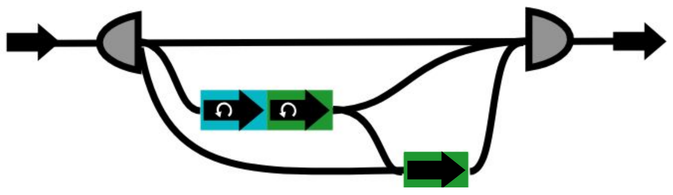
1. Find seeds in graph



2. Guess orientiantation and order of subsections



3. Make zip code tree with artificial children

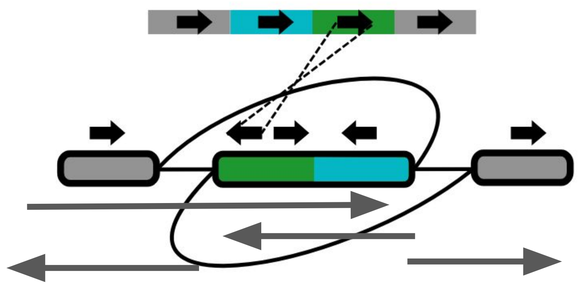


Algorithm

New exact algorithm

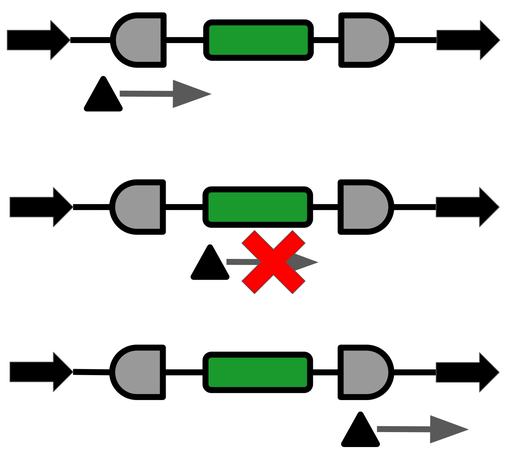
- Subsections may have **arbitrary orientation** and **order**
- Each subsection is traversed **once in each direction**

… but wouldn't that lead to a runtime explosion?

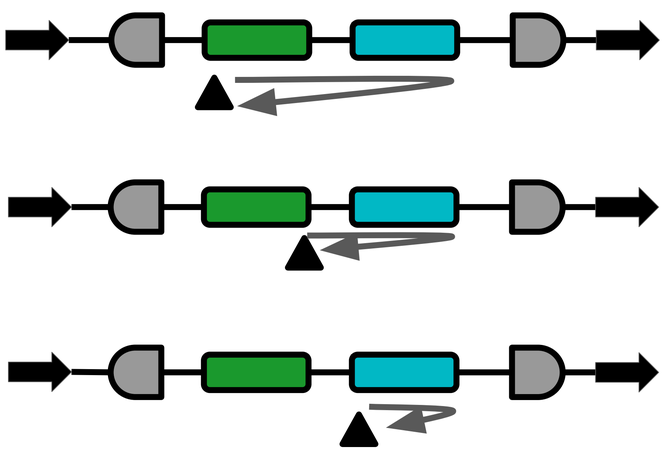


Optimizations

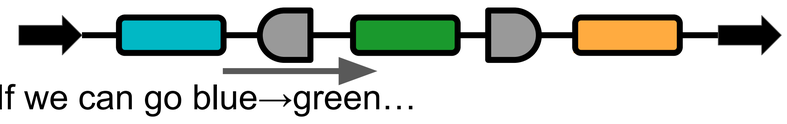
Jumping



Avoid duplicate traversals

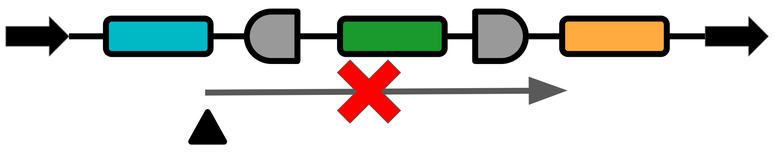


Never enter a cyclic snarl



If we can go blue→green…

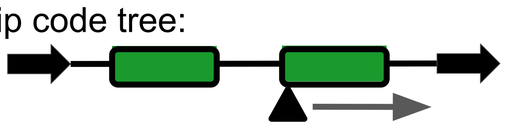
… then we'd get a green→blue connection later



Duplicate minimizers

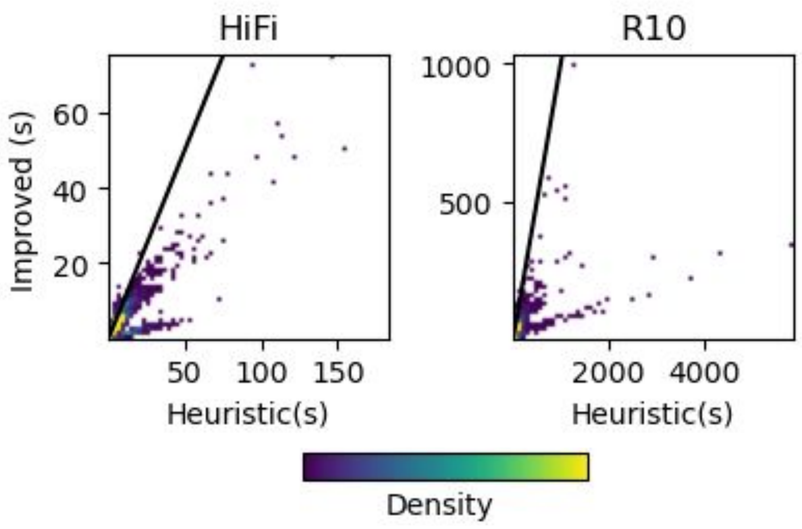
Old zip code tree:

New zip code tree:



Results

Faster alignments



(slightly) Better alignments

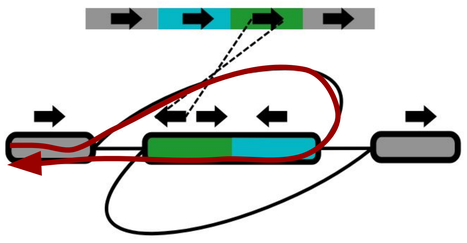
	Old correct	New correct
HiFi	925,084	925,086
R10	925,525	925,545

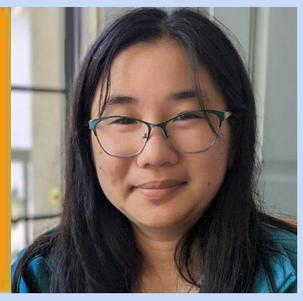
	Old softclips	New softclips
HiFi	0.153%	0.152%
R10	0.743%	0.722%

Giraffe is under active development

- **Recombination-aware** mode (Davide Monti)
- --diploid-mode for **personalized assemblies** (Sagorika Nag)
- Simplified chaining from two steps to one (Adam Novak)
- Improvements to **distance index** creation (Zia Truong)

Plus some final edge cases for the cyclic snarl handling.



Acknowledgments

Thanks to vg team for support! Also, many figures are taken from Chang *et al.* 2025 (bioXiv), of which I am a co-author.